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EECS 148 Homework 4

1. IP Addressing

Chapter 4, P13

Subnet 1: 223.1.17.0/26

Subnet 2: 223.1.17.128/25

Subnet 3: 223.1.17.64/28

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1. Home addresses: 192.168.1.1, 192.168.1.2, 192.168.1.3
Router interface: 192.168.1.4

2. NAT Translation Table

WAN Side	LAN Side
24.34.112.235, 5001	192.168.1.1, 3345
24.34.112.235, 5002	192.168.1.1, 3346
24.34.112.235, 5003	192.168.1.2, 3347
24.34.112.235, 5004	192.168.1.2, 3348
24.34.112.235, 5005	192.168.1.3, 3349
24.34.112.235, 5006	192.168.1.3, 3350

2. BGP

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W's view X's view

From X's view, there is no link between A and C because X does not get an advertised route that uses that link to get to W or Y.

3. Routing Algorithms

(a) Route: H_1 -A-B-C-D-E- H_2

Step	S	d(A), p(A)	d(B), p(B)	d(C), p(C)	d(D), p(D)	d(E), p(E)	d(H ₂), p(H ₂)
0	H ₁	10, H ₁	∞	∞	∞	∞	∞
1	H ₁ A		20, A	∞	60, A	∞	∞
2	H ₁ AB			40, B	60, A	100, B	∞
3	H ₁ ABC				50, C	100, B	∞
4	H ₁ ABCD					90, D	∞
5	H ₁ ABCDE						100, E
6	H ₁ ABCDEH ₂						

*cost for each link is transmission delay + propagation delay in milliseconds

(b) No. This route is not the same as the minimum number of hops.

(c) Initial Tables

Node A (B,D) | Node B (A,C,D,E) | Node C (B,D,E) | Node D (A,B,C,E) | Node E(B,C,D)

Dest	Via	Delay	Dest	Via	Delay	Dest	Via	Delay	Dest	Via	Delay	Dest	Via	Delay
A	A	0	A	-	∞	A	-	∞	A	-	∞	A	-	∞
B	-	∞	B	B	0	B	-	∞	B	-	∞	B	-	∞
C	-	∞	C	-	∞	C	C	0	C	-	∞	C	-	∞
D	-	∞	D	-	∞	D	-	∞	D	D	0	D	-	∞
E	-	∞	E	-	∞	E	-	∞	E	-	∞	E	E	0

Tables after first exchange

Node A (B,D) | Node B (A,C,D,E) | Node C (B,D,E) | Node D (A,B,C,E) | Node E(B,C,D)

Dest	Via	Delay	Dest	Via	Delay	Dest	Via	Delay	Dest	Via	Delay	Dest	Via	Delay
A	A	0	A	B	10	A	-	∞	A	D	50	A	-	∞
B	A	10	B	B	0	B	C	20	B	D	40	B	E	80
C	-	∞	C	B	20	C	C	0	C	D	10	C	E	60
D	A	50	D	B	40	D	C	10	D	D	0	D	E	40
E	-	∞	E	B	80	E	C	60	E	D	40	E	E	0

Tables after second exchange

Node A (B,D) | Node B (A,C,D,E) | Node C (B,D,E) | Node D (A,B,C,E) | Node E(B,C,D)

Dest	Via	Delay	Dest	Via	Delay	Dest	Via	Delay	Dest	Via	Delay	Dest	Via	Delay
A	A	0	A	B	10	A	B	30	A	D	50	A	B	90
B	A	10	B	B	0	B	C	20	B	C	30	B	E	80
C	B	30	C	B	20	C	C	0	C	D	10	C	D	50
D	A	50	D	C	30	D	C	10	D	D	0	D	E	40
E	B	90	E	B	80	E	D	50	E	D	40	E	E	0

Tables after third exchange

Node A (B,D) | Node B (A,C,D,E) | Node C (B,D,E) | Node D (A,B,C,E) | Node E(B,C,D)

Dest	Via	Delay	Dest	Via	Delay	Dest	Via	Delay	Dest	Via	Delay	Dest	Via	Delay
A	A	0	A	B	10	A	B	30	A	B	40	A	B	90
B	A	10	B	B	0	B	C	20	B	C	30	B	C	70
C	B	30	C	B	20	C	C	0	C	D	10	C	D	50
D	C	40	D	C	30	D	C	10	D	D	0	D	E	40
E	B	90	E	D	70	E	D	50	E	D	40	E	E	0

Tables after fourth exchange

Node A (B,D) | Node B (A,C,D,E) | Node C (B,D,E) | Node D (A,B,C,E) | Node E(B,C,D)

Dest	Via	Delay	Dest	Via	Delay	Dest	Via	Delay	Dest	Via	Delay	Dest	Via	Delay
A	A	0	A	B	10	A	B	30	A	B	40	A	B	80
B	A	10	B	B	0	B	C	20	B	C	30	B	C	70
C	B	30	C	B	20	C	C	0	C	D	10	C	D	50
D	C	40	D	C	30	D	C	10	D	D	0	D	E	40
E	D	80	E	D	70	E	D	50	E	D	40	E	E	0

(d) Yes. The algorithm converges.

Tables at steady state

Node A (B,D) | Node B (A,C,D,E) | Node C (B,D,E) | Node D (A,B,C,E) | Node E(B,C,D)

Dest	Via	Delay	Dest	Via	Delay	Dest	Via	Delay	Dest	Via	Delay	Dest	Via	Delay
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A	A	0	A	B	10	A	B	30	A	B	40	A	B	80
B	A	10	B	B	0	B	C	20	B	C	30	B	C	70
C	B	30	C	B	20	C	C	0	C	D	10	C	D	50
D	C	40	D	C	30	D	C	10	D	D	0	D	E	40
E	D	80	E	D	70	E	D	50	E	D	40	E	E	0

Tables after links AB and BE fail: first exchange

Node A (B,D) | Node B (A,C,D,E) | Node C (B,D,E) | Node D (A,B,C,E) | Node E(B,C,D)

Dest	Via	Delay	Dest	Via	Delay	Dest	Via	Delay	Dest	Via	Delay	Dest	Via	Delay
A	A	0	A	B	∞	A	B	∞	A	B	∞	A	B	∞
B	A	∞	B	B	0	B	C	20	B	C	30	B	C	70
C	B	30	C	B	20	C	C	0	C	D	10	C	D	50
D	A	50	D	C	30	D	C	10	D	D	0	D	E	40
E	D	80	E	D	70	E	D	50	E	D	40	E	E	0

Tables after links AB and BE fail: second exchange

Node A (B,D) | Node B (A,C,D,E) | Node C (B,D,E) | Node D (A,B,C,E) | Node E(B,C,D)

Dest	Via	Delay	Dest	Via	Delay	Dest	Via	Delay	Dest	Via	Delay	Dest	Via	Delay
A	A	0	A	D	80	A	D	60	A	D	50	A	D	90
B	D	90	B	B	0	B	C	20	B	C	30	B	C	70
C	B	30	C	B	20	C	C	0	C	D	10	C	D	50
D	A	50	D	C	30	D	C	10	D	D	0	D	E	40
E	D	90	E	D	70	E	D	50	E	D	40	E	E	0

***delay is in milliseconds for question 3, parts (c) and (d)